

Visible/Near Infrared (VIS/NIR) spectroscopy as an optical sensor for evaluating olive oil quality

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ABSTRACT

The quality of olive oils can be tested by chemical ways, and these methods are time consuming and very expensive. Therefore, the aim of this study is focused on reducing both time and cost, by using Visible/Near Infrared (VIS/NIR) spectroscopy (*i.e.* 400–1100 nm) absorbance to evaluate the quality of olive oil. Thus, 48 olive oil samples were collected from different governorates in Palestine and the samples covered the four quality groups (*i.e.* extra virgin, virgin, ordinary virgin and lampante). Chemical analysis was carried out, and then spectra were acquired by VIS/NIR spectroscopy. Spectral data was analyzed by principal component analysis (PCA). The relation between spectra and chemical data were evaluated by partial least squares (PLS). It was found the PCA was able to distinguish between four quality groups, also it was found that the VIS/NIR spectroscopy was able to model some quality parameters (*i.e.* acidity and peroxide) with good accuracy. The PLS correlation coefficients for acidity and peroxide were 0.88 and 0.86 for validation sets, respectively. The ratios of standard error of performance to standard deviation (RPD) values were 3.08 and 2.25 for validation set for acidity and peroxide, respectively. The results showed that it was possible to model/predict acidity and peroxide parameters with good accuracy. Moreover, it was also possible to distinguish between different quality groups of olive oil.

1. Introduction

The olive tree (*Olea europaea* L.) is one of the most ancient trees cultivated in the Mediterranean regions. Olive trees play an essential role in the social and economic life of this region. It is one of the oldest agricultural trees cultivated over large areas in Palestine (Qutub et al., 2010). It has a deep cultural significance as a symbol of traditional society and ties to the land. The number of olive trees in Palestine is exceeding 10 million (about 67.3% of horticultural trees) and occupying more than 50% of the agricultural area (Palestinian Central Bureau of Statistics, 2018).

The agriculture sector in Palestine accounts for nearly 25% of GDP. The olive sector is an important element of the Palestinian economy and estimates suggest that about 100,000 families depend to some extent upon the olive harvest for their livelihoods. The trees comprise one of the main sources of income, reaching its contribution in good years to about 13% of the annual agricultural production (World Bank, 2014).

Olive oil is the primary fat used by the Palestinian population, and most production is locally consumed. Significant exports take place

only in high production years, when there is a large surplus above local needs. Also, the international markets ask for high quality standards (extra Virgin mainly and some virgin olive oil). It is impossible to accurately determine the volume of exports because they are cyclical and many occur informally to Israel or as gifts to family members in Jordan or Arab Gulf countries. But in very good years, as much as 20–30% of the total production might be exported out of Palestine.

In the last 30 years, there has been an emphasis on the quality, safety and nutritional proprieties of food products and thus an increasing interest by food producers to reveal frauds and to assure consumers regarding the quality and provenance of their products. In particular, olive oil has become more popular as a healthy food in contrast with the rapid decrease in world demand for animal fats in all their forms (Casale and Simonetti, 2014).

The olive oil should have a high quality to be consumed and exported. However, sometimes it is exposed to degradation of quality due to different parameters or adulteration, and this harm the consumption, exporting and reputation of the Palestinian olive industry. Consumers usually rely on color, appearance and ratings to make a decision before

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buying olive oil, so it is difficult for the consumer to detect adulteration, because of the similarity between the physical properties of virgin olive oil and other oils that maybe are mixed with olive oil. Adulteration is defined as altering a product by substituting components of high monetary value with cheaper ingredients and claiming that the product is genuine (Vanstone et al., 2018).

Testing the quality of olive oil using traditional ways (e.g. chemical analysis) is costly, laborious and time-consuming. An alternative is using optical sensors, like Visible/Near Infrared (VIS/NIR) spectroscopy, for checking the quality, adulteration and geographical origin of olive oil.

VIS/NIR spectroscopy is a non-destructive method for measuring chemical compounds in heterogeneous products. In addition, it is a low cost, time saving method and it is environmentally friendly, because it is not using harmful chemicals, such as bases and others that are used by traditional methods to check the quality of olive oil (Giovenzana et al., 2018). This technique is used to determine the properties of foods for obtaining the characteristics and chemical composition. VIS/NIR spectroscopy has emerged in the food sector as a rapid and very useful tool for such purposes as quality control, process monitoring and estimating shelf-life (Tahir et al., 2019).

VIS/NIR spectra signals need interpretation. Multivariate data analysis (MVDA), also called chemometrics, is an essential part of the acquired spectra to achieve useful and meaningful results of revealed spectra. MVDA, which involves pattern recognition (e.g. principal component analysis (PCA)), and regression techniques (e.g. partial least squares (PLS)), is therefore required to extract the information hidden in the spectra (Abu-Khalaf, 2015; Zaid et al., 2020). PCA is a qualitative method, which shows the linear relationship between different samples and variables, and the possibility of classifying different groups during the experiments, through investigating scores and loading plots of different treatments during experiment period. PLS is the most extensively used chemometrics method in the VIS/NIR spectroscopy field. PLS is used as a quantitative method for the establishment of the regression/prediction models of reference (e.g. chemical) parameters. It consists of relating two data matrices, X and Y, by a linear multivariate model and modeling their structure (Farres et al., 2019).

There are many uses for VIS/NIR spectroscopy in the food field (Porep et al., 2015). In addition, it was used to check the adulteration in olive oil (Vanstone et al., 2018). There are several previous studies in which the VIS/NIR technique with range of 400–2500 nm was used to check the quality of olive oil, as it was applied at the local level for olive varieties in some countries such as Spain, Italy and others (Jolayemi et al., 2017; Vanstone et al., 2018). However, in this study the range was 400–1100 nm. Taking into consideration that the 400–1100 nm sensors are cheaper than sensors for longer wavelengths, and if this type of sensor can be proven to perform adequately, this can facilitate the commercial implementation of the technology.

NIR spectroscopy together with chemometrics data analysis has shown significant potential for the determination of useful compounds present in olive oil (Mahesar et al., 2019), and non-destructive assessment of chemical quality parameters in olive oil (Garrido-Varo et al., 2017). Moreover, NIR was used to check the quality of olive oil such as free acidity, peroxides value, K270 and K232 and ΔK. The author built a PLS model with a correlation coefficient (R^2) of 0.84 to measure the free fatty acid. Moreover, he was able to examine the peroxide in olive oil by using the strain on the wave range of 1100–2500 nm (Cayuela, 2017a).

Valli et al. (2016) have used PLS calibration model for measurement acidity in olive oil samples by NIR with range of 750–2500 nm, and they obtained a R^2 up to 0.970. Also, Vallone et al. (2019) have excellent performance in predicting table olives color (i.e. $R^2 = 0.96$) by VIS/NIR with a range of 600–1000 nm. Karabagias et al. (2019) tested some quality parameters for 37 olive oil samples, like acidity, peroxide values, K232, K270 and ΔK indices, and the parameters were exposed to linear discriminant analysis (LDA) providing a correct grouping rate of 91.9% and 81.1% using the original and cross-validation methods,

respectively. Giovenzana et al. (2018) were successful in using a VIS/NIR with a wavelength of 400–1650 nm to study the oil content during the olive processing, where R^2 and ratio of standard error of performance to standard deviation (RPD) values were 0.77 and 3.00, respectively.

Borras et al. (2015) used NIR to build a partial least squares discriminant analysis (PLS-DA) model to distinguish between extra virgin and lower quality olive oils with a range of 1666–2500 nm. PLS-DA models were able to discriminate between defective and high-quality oils with predictive abilities around 87%.

This study aims to serve the local society and introducing the usage of VIS/NIR spectroscopy (i.e. sensor technology) in the range of 400–1100 nm, in the olive sector and contributing in a fast checking of the olive oil quality in Palestine. To our best knowledge, this study is considered one of the first studies to use the range 400–1100 nm for olive oil quality parameters quantifications and group classification, whereas previous studies had used range above 1100 nm for this purpose.

2. Materials and methods

2.1. Olive oil samples

Different olive oil samples (i.e. 48 samples) were collected from different southern, middle and northern governorates in Palestine. They were collected from Hebron, Ramallah, Salfeet, Nablus, Qalqilia, Tulkarm and Jenin in olive season 2018. The samples were collected from Palestinian farmers and farmer cooperatives.

Samples were stored in dark tightly closed bottles at 17 °C until and during chemical and spectra measurements, since exposure to oxygen and light lead to oxidation of oil and thus affect the results. Also, storing the samples at temperatures less than 12 °C causes the freezing of the oil, and a temperature higher than 17 °C leads to rapid degradation of the oil. It is necessary to measure all samples at the same temperature, because the difference in the temperature of samples during measurement affects the absorption of the wavelength of the NIR spectrum so samples are measured at room temperature.

2.2. Chemical analysis of olive oil

Seven chemical parameters were tested. They were:

2.2.1. Acidity test

About 7 g of dried, filtered and well-mixed oil were weighed into 250 ml Erlenmeyer flask. Then, 50 ml of 96% ethanol was taken and neutralized with 0.1 N aqueous sodium hydroxide (NaOH) solution in the presence of 2.0 ml phenolphthalein solution to produce faint permanent pink. The neutralized ethanol was added to the oil in the flask. The whole mixture was vigorously shaken and boiled on a hot plate for two minutes. Thereafter, the flask contents were titrated with 0.1 N aqueous NaOH solution until permanent faint pink and appear persisting for one minute. Results could be reported as acidity percentage - free fatty acids (FFA) contents, which mean that the percent of FFA expressed as a certain fatty acid of specified molecular weight according to the type of oil under investigation. Normally oleic acid with a molecular weight of 282 is taken. In several cases, an average molecular weight, more appropriate to the nature of the oil, is used. It is calculated according to the following equation (Eq. (1)) (Cayuela, 2017b):

$$\text{Acidity\%} = (N * V * G * 100) / (1000 * W) \quad (1)$$

Where:

N: Normality of NaOH aqueous solution.

V: Milliliters of NaOH aqueous solution required to neutralize fatty acids present in the used sample.

G: is the molecular weight of the appropriate fatty acid.
W: grams of the sample used in the test.

2.2.2. Peroxide value test

About 5 g of oil was weighed into a 250 ml glass-stoppered conical flasks. The acetic acid-Chloroform solution (3:1 V/V) was added (30 ml) with swirling to dissolve oil completely. The saturated potassium iodide solution (1.0 ml) was added. The flask was quickly stoppered and let to stand with occasional shaking for 1 min in the dark. Thereafter, 30 ml of freshly boiled and cooled water was added and about 0.5 ml of starch solution was added and flask contents were titrated with 0.01 N sodium thiosulfate solution ($\text{Na}_2\text{S}_2\text{O}_3$) with vigorous shaking until the dark yellow or dark blue color had almost gone from the chloroform layer. A blank determination was conducted in the same way without using oil.

Peroxide value is the number of milliequivalents of peroxide found in 1000 g of the sample according to this formula (Eq. (2)) (Cayuela, 2017a).

$$\text{Peroxide value} = ((V1 - V2) * N * 100)/W \quad (2)$$

Where:

V1: Volume (in ml) of solution thiosulfate used in test.
V2: Volume (in ml) of sodium thiosulfate used in blank test
N: Normality of sodium thiosulfate solution.
W: Weight (in g) of sample.

2.2.3. K-values test

A 0.25 g of the oil sample was weighed and placed in 25 ml graduated flask, to which was added a solvent of *iso*-octane or cyclohexane. Samples were placed in a cuvette and absorption measured at wavelengths between 232 and 276 nm with the solvent as reference. Where *iso*-octane was used as a solvent, maximum wavelengths of 268 nm and 232 nm were used. For cyclohexane, the maximum wavelengths were 270 nm and 232 nm (Karabagias et al., 2019). K232, K270, K266, K274 and ΔK were measured.

Based on the results of the chemical examination and according to the International Olive Council (IOC) (Valli et al., 2016): 20 of our samples were graded as virgin olive oil (VOO), 9 as extra virgin olive oil (EVOO), 15 as ordinary virgin olive oil (OVOO) and 4 as lampante olive oil (LOO). The grading system used in IOC is shown in Table 1.

2.3. VIS/NIR spectroscopy

A USB2000 + XR1-ES (Ocean Optics, USA) spectrometer was used as VIS/NIR sensor, with a halogen lamp HL-2000-FHSA as a light source. Optical fibers were used for conducting the absorbance light through to the samples. All data were recorded using Spectra Suite Software (Ocean Optics, USA). Samples are placed in disposal cuvette with inner diameter 3 mm. The integration time was 400 ms, the scans to average value was 90 and boxcar width was 20.

2.4. Chemometrics

Once samples measurements are completed, identification and quantification models can be created from the spectra using computer software. The spectra are exported from Microsoft excel program, and

digitally manipulated using The Unscrambler X (version 10.4, CAMO Software AS, Norway) multivariate data analysis software.

2.5. Data analysis

PCA was used initially to examine the possible classification of the four grades of olive oil samples (*i.e.* extra virgin, virgin, ordinary virgin and lampante) for VIS/NIR region with a range 400–1100 nm with full cross validation.

For studying the possibility of using spectra to predict chemical parameters of olive oil, PLS-1 (one reference Y was used) calibration models were established. Spectral data were used as X-matrix and chemical measurements were used as Y-matrix in the PLS-1 model. Different pre-processing methods of the raw spectra were performed before building PLS models. For example, standard normal variate (SNV) (Barnes et al., 1989), Savitzky-Golay smoothing (SG) and Multiplicative signal correction (MSC) (Geladi et al., 1985) were used. Test set validation was used. About two-third of the samples was used as a calibration set (*i.e.* 34 samples), and one-third was used as a validation set (*i.e.* 14 samples).

2.6. Model performance

Different parameters were used to evaluate PLS model. The coefficient of correlation in calibration (R_{cal}^2) and validation (R_{val}^2) models were used to estimate model accuracy. These value defined how well the achieved data point fit the line of the PLS regression model. These values range from 0 to 1. Moreover, the root mean squares error of calibration and prediction (RMSE_{cal}) and (RMSE_{val}) were used.

To state the acceptance or rejection of the PLS model, RPD and relative error (RE) were used. RPD is defined as follows (Eq. (3)):

$$\text{RPD} = \text{SD}/\text{RMSE} \quad (3)$$

Generally, different studies suggest that a PLS model with an RPD value less than 2.0 doesn't provide a sufficient prediction, while RPD value between 2.0 and 3.0 considered as a good PLS model, whereas the model with RPD more than 3.0 gives an excellent prediction (Beghi et al., 2018; Escuredo et al., 2018; Porep et al., 2015).

RE is calculated as in the following relation (Eq. (4)):

$$\text{RE}(\%) = \text{RMSE}/((\text{Maximum} + \text{Minimum})/2) \quad (4)$$

A PLS model can primarily be accepted when it has high R^2 , low number of principal components (PCs) used in the model, low RME, RPD is greater than 2.0 and lower difference between both calibration and validation sets (Abu-Khalaf et al., 2004; Abu-Khalaf and Iversen, 2007; Abu-Khalaf et al., 2013).

3. Results and discussion

3.1. Descriptive statistics

The descriptive statistics of chemical quality parameters according to olive oil groups is shown in Table 2. Mean, standard deviations (SD), maximum and minimum values and range (*i.e.* maximum value – minimum value) obtained from seven quality parameters (*i.e.* acidity, peroxide, K232, K270, K266, K274 and ΔK) are shown in Table 3.

Table 1

Olive oil grading system according to the International Olive Council (IOC).

Olive oil grades	Acidity (%)	Peroxide (mEq O ₂ /kg)	K232 absorbance UV	K270 absorbance UV	ΔK
Extra virgin	≤ 0.8	≤ 20	< 2.5	< 0.22	< 0.01
Virgin	0.8–2.0	≤ 20	< 2.6	< 0.22	< 0.01
Ordinary Virgin	2.0–3.3	≤ 20	< 2.6	< 0.22	< 0.01
Lampante	≥ 3.3	≥ 20	> 2.6	> 0.22	> 0.01

Table 2

Descriptive statistical of quality parameters for four groups of 48 olive oil samples measured in the research using chemical analysis.

Groups		Acidity (%)	Peroxide (mEq O ₂ /kg)	K232 (UV)	K270 (UV)	K266 (UV)	K274 (UV)	ΔK (UV)
Virgin	Mean	1.71	15.29	1.95	0.19	0.20	0.18	0.00
	SD	0.94	3.39	0.27	0.06	0.07	0.06	0.00
	Minimum	0.60	9.20	1.52	0.10	0.10	0.07	−0.02
	Maximum	2.00	19.00	2.54	0.20	0.40	0.36	0.00
	Range	1.40	9.80	1.02	0.10	0.30	0.28	0.02
Extra virgin	Mean	1.45	16.57	2.12	0.24	0.24	0.23	0.00
	SD	0.98	1.59	0.13	0.05	0.06	0.06	0.01
	Minimum	0.36	14.40	1.95	0.17	0.17	0.16	−0.01
	Maximum	0.80	19.00	2.30	0.19	0.33	0.33	0.01
	Range	0.44	4.60	0.35	0.02	0.17	0.17	0.02
Ordinary virgin	Mean	1.87	17.17	2.01	0.23	0.24	0.23	0.00
	SD	1.11	4.38	0.20	0.05	0.06	0.05	0.00
	Minimum	0.60	12.20	1.51	0.14	0.15	0.13	−0.01
	Maximum	3.30	18.00	2.24	0.22	0.41	0.35	0.00
	Range	2.70	5.80	0.74	0.08	0.25	0.22	0.01
Lampante	Mean	1.43	12.25	2.07	0.25	0.26	0.25	0.00
	SD	0.39	3.24	0.22	0.09	0.09	0.09	0.00
	Minimum	1.12	10.00	1.79	0.15	0.16	0.15	0.01
	Maximum	3.30	25.00	2.33	0.37	0.38	0.38	0.10
	Range	2.18	15.00	0.53	0.22	0.22	0.23	0.09

SD: standard deviation, Range: Maximum – Minimum.

UV: absorbance UV.

Table 3

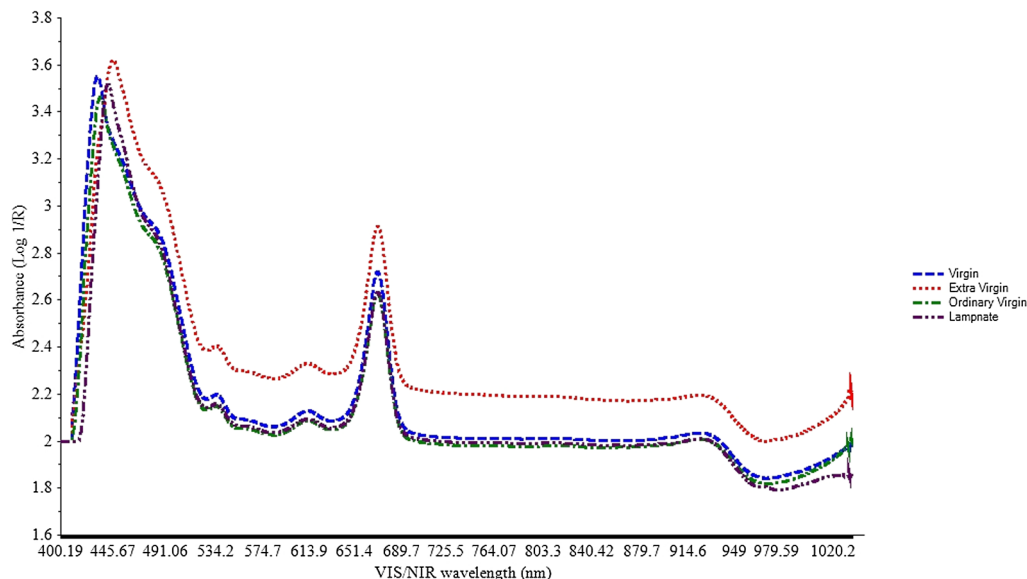
Descriptive statistics of seven quality parameters for tested olive oil samples (48 samples).

Quality Parameters	Mean	SD	Minimum	Maximum	Range
Acidity (%)	1.66	0.92	0.36	3.30	2.94
Peroxide (mEq O ₂ / kg)	15.52	3.65	8.00	25.00	17.00
K232 (absorbance UV)	2.01	0.23	1.51	2.54	1.04
K270 (absorbance UV)	0.22	0.06	0.09	0.37	0.28
K266 (absorbance UV)	0.23	0.07	0.10	0.41	0.30
K274 (absorbance UV)	0.21	0.07	0.07	0.38	0.30
ΔK (absorbance UV)	0.00	0.00	−0.02	0.01	0.03

SD: standard deviation, Range: Maximum – Minimum.

3.2. Spectral characterization:

Spectral curve representing the four groups of olive oil in the wavelength range of 400–1100 nm is shown in Fig. 1. The depressions and peaks in spectra showed the strong and weak absorbance characteristics

**Fig. 1.** VIS/NIR (400–1100 nm) spectral curves representing olive oil groups.

of the samples. Data pre-treatment was needed to minimize noising and scattering effects. Firstly, pre-processed using SNV (Barnes et al., 1989) and MSC (Geladi et al., 1985) to delete slope variation and to correct for scattering effects, then using SG smoothing to prevent increasing the noise, were carried out (Beghi et al., 2018; Farres et al., 2019).

Six bands (i.e. peaks at 450.5, 532, 601, 673, 932 and 977 nm) in the VIS/NIR region have been observed. Our finding showed that extra virgin and virgin had higher absorption than lampante and ordinary virgin in almost all the range of spectra.

3.3. PCA

PCA was able to separate the samples according to the olive oil grade (i.e. extra virgin, virgin, ordinary virgin and lampante) using VIS/NIR range of 400–1100 nm as shown in Fig. 2. Two PCs explained 92% of the total variance. Such results are in agreement with the previous study with range above 1100 nm, where only two PCs were able to explain such results (Borras et al., 2015).

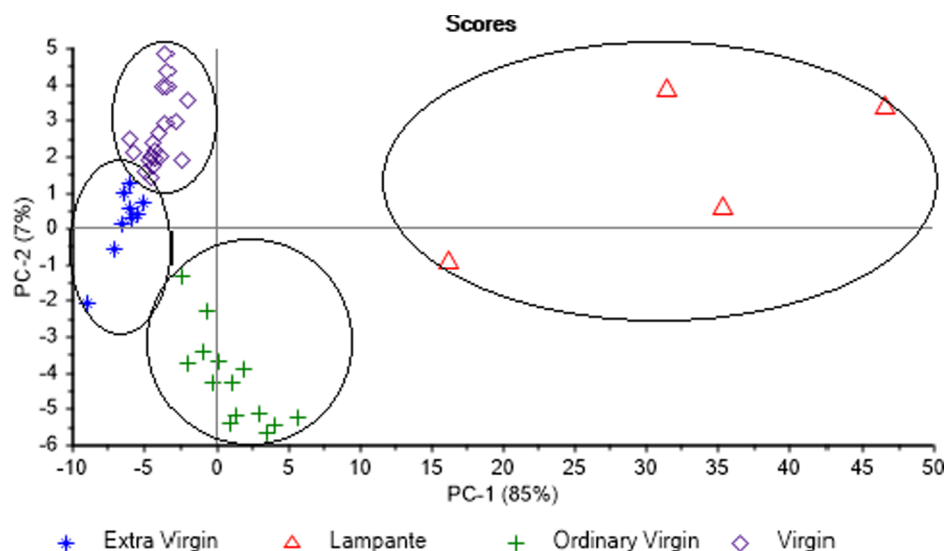


Fig. 2. PCA scores plot of VIS/NIR region with a range of 400–1100 nm for all samples from different groups of olive oil samples.

3.4. PLS prediction models

PLS models were established for the seven quality parameters. However, it was found that only acidity and peroxide had well PLS models according to model performance parameters. The others (*i.e.* K232, K274, K266, K270 and ΔK) had badly PLS models (*i.e.* low R^2 , high RME, low RPD and big differences between both calibration and validation sets results) (data not are shown).

PLS models of calibration and validation for acidity and peroxide are shown in Figs. 3–6. PLS model results are shown in Table 4.

For acidity PLS model, the correlation coefficients for calibration and validation sets were both 0.88. RMSE had almost the same value (*i.e.* around 0.35). RPDs were 2.78 and 3.08 for calibration and validation sets, respectively. The relative errors were less than 20% for both sets. Moreover, the model has a high slope (*i.e.* greater than 0.96 in both sets).

Peroxide PLS model showed also a good performance. The correlation coefficients for calibration and validation sets were 0.83 and 0.86, respectively. RMSE had almost the same value (*i.e.* around 1.6). RPDs had almost the same value (*i.e.* 2.2) in both sets. The relative errors were equal or less than 10% for both sets. Moreover, the model showed a high slope (*i.e.* greater than 0.88 in both sets).

Both models can be considered as acceptable models, since they showed a good performance, and there is a little difference between results of calibration and validation sets. These results coincided with previous studies, where VIS/NIR spectroscopy was used to predict olive oil acidity by using PLS model (Giovenzana et al., 2018).

The novelty of this environmentally friendly technique can affect the quality control of olive oil in both small (*i.e.* farmers) and big scales (*i.e.* agricultural exporting companies). Acidity and peroxide are the main chemical analysis that should conduct on the olive oil to evaluate the quality grade of the olive oil. These analyses are currently done by

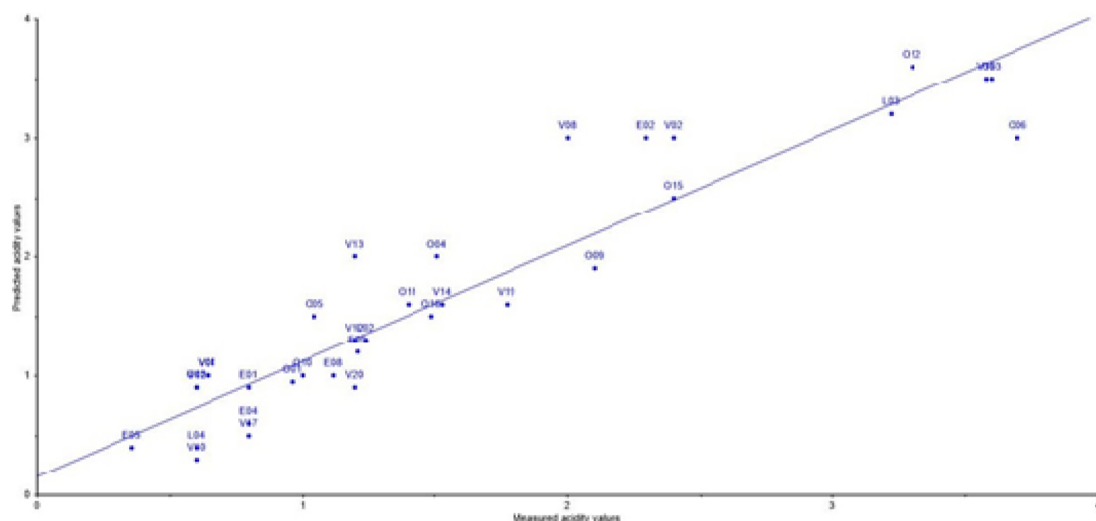


Fig. 3. PLS model for acidity calibration set. Samples named according to their grading group: E: extra virgin, V: virgin, OV: ordinary virgin and L: lampante.

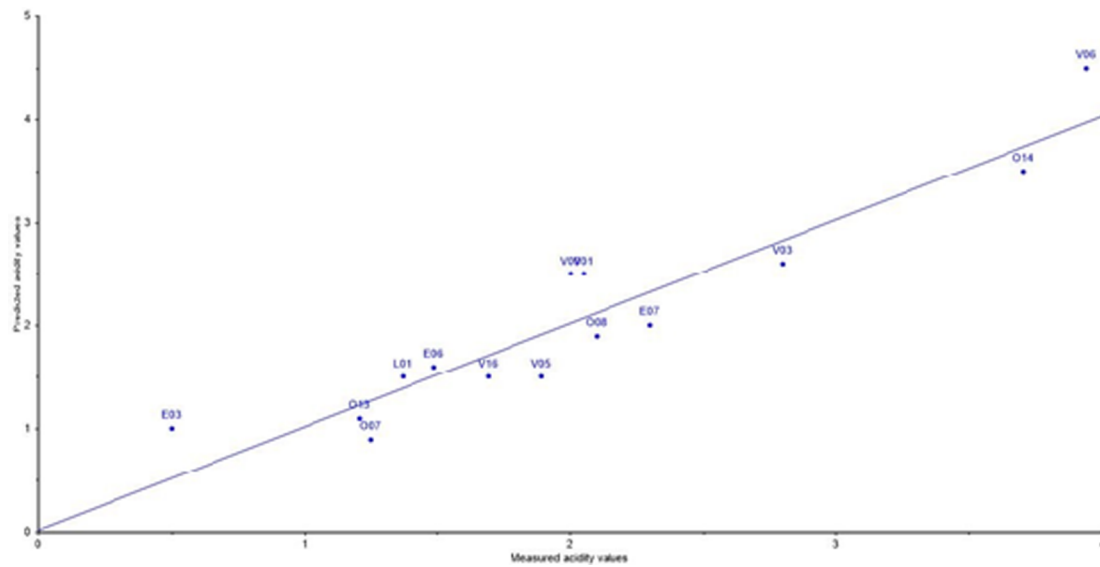


Fig. 4. PLS model for acidity validation set. Samples named according to their grading group: E: extra virgin, V: virgin, OV: ordinary virgin and L: lampante.

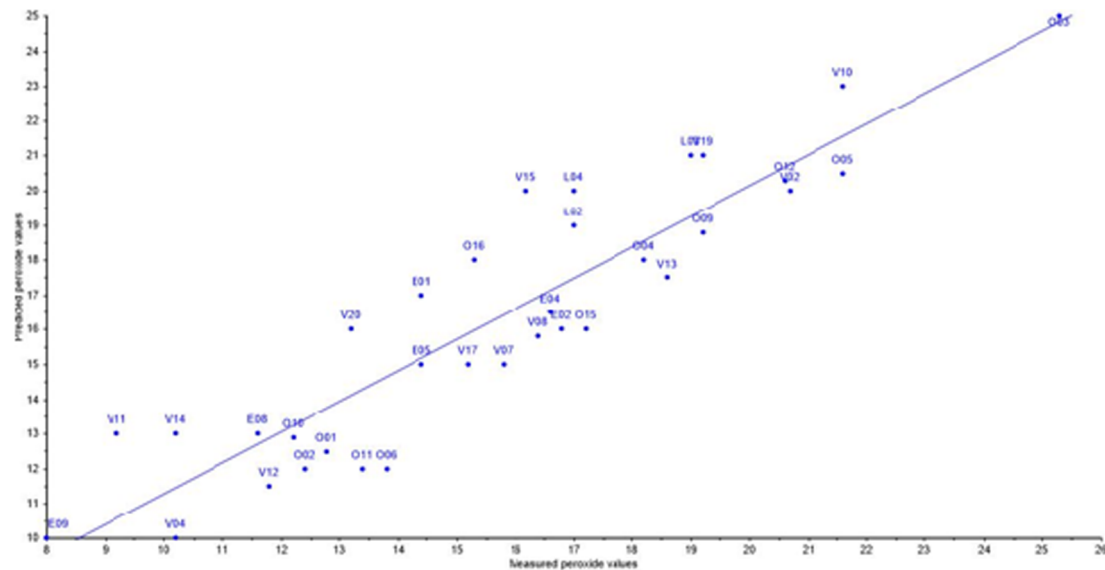


Fig. 5. PLS model for peroxide calibration set. Samples named according to their grading group: E: extra virgin, V: virgin, OV: ordinary virgin and L: lampante.

destructive methods, based mainly on titration with chemical reagents according to IOC's procedure. The titration methods are destructive, where the samples are destroyed during analysis, also these methods are time-consuming and need trained and qualified technicians to do them, besides that, they are costly methods. In Palestine, the ownership of the olive orchards is very small and distributed in different areas of the country, so the production for each family or each ownership is very small, for that the titration methods are too costly per kilogram of olive oil and too time-dependent to collect samples from wide range of locations and send them to the laboratory and analyze them. Moreover, the method can be used in the quality control procedure in agricultural exporting companies. The NIR non-destructive method is a novel technique to analyze acidity and peroxide of olive oil directly without collecting samples, so the analysis can be carried out immediately on the olive oil on time and the decision of the olive oil quality grade will be taken without waiting for the laboratory results. The NIR method is non time-consuming and not costly. Furthermore, there is a possibility

of using a portable NIR, which can be used for quality control measurement in fields and borders. This could be a novel technique, saving time and efforts.

4. Conclusion

In conclusion, the findings of this study showed that VIS/NIR spectroscopy (400–1100 nm) was satisfactory to predict some quality parameters, like acidity and peroxide for olive oil with a high accuracy. The PLS models with two PCs had shown high R^2 , low RMSE, RPD higher than 2.0 and a relatively low RE. Moreover, the VIS/NIR could differentiate between the four olive oil groups (extra virgin, virgin, ordinary virgin and lampante). Besides, the results open a wide door for the potential of using a portable VIS/NIR technique in the olive oil industry. However, further research is needed in this field.

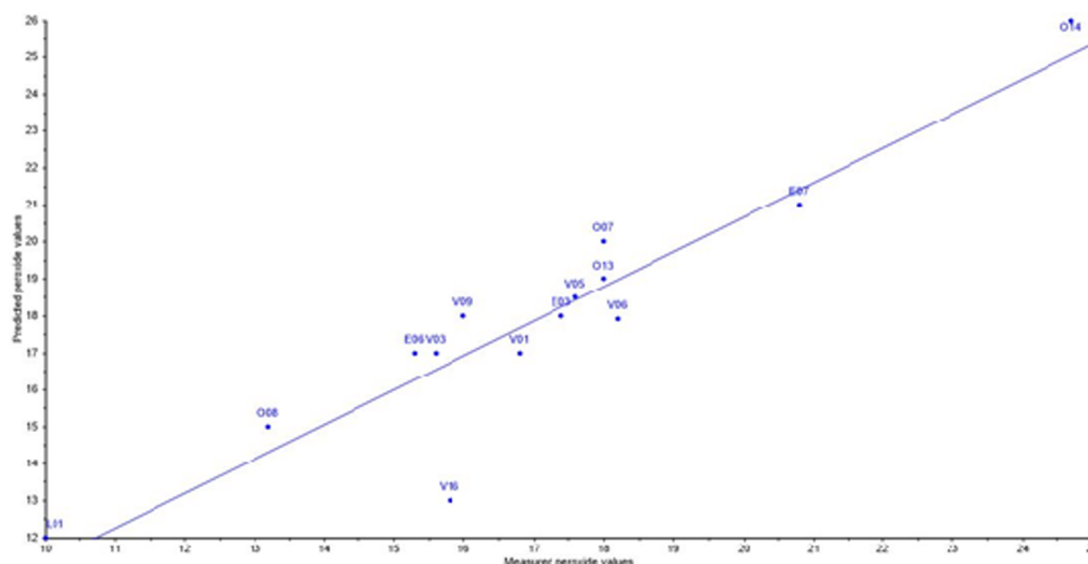


Fig. 6. PLS model for peroxide validation set. Samples named according to their grading group: E: extra virgin, V: virgin, OV: ordinary virgin and L: lampante.

Table 4

PLS-1 models' results for the calibration (cal) and validation (val) sets for acidity and peroxide of olive oil. Calibration and validation sets have 34 and 14 samples, respectively.

Quality parameter	Acidity	Peroxide
R^2_{cal}	0.88	0.83
$RMSE_{cal}$	0.36	1.71
RPD_{cal}	2.78	2.23
RE_{cal}	19%	10%
$Slope_{cal}$	0.97	0.89
R^2_{val}	0.88	0.86
$RMSE_{val}$	0.34	1.51
RPD_{val}	3.08	2.25
RE_{val}	12%	8%
$Slope_{val}$	1.00	0.93

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2020.105445>.

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